

ngspice commands — a worked reference

Every command in this ngspice-46 + OpenVAF-r build, with netlists and plots

Ngspice + OpenVAF Enhancements project

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Contents

ngspice commands — a worked reference	1
1. How commands are run	1
2. Analyses	2
3. RF and periodic steady-state	8
4. Signal processing and measurement	11
5. Vectors, variables and expressions	12
6. Plotting and data export	13
7. Circuit and device management	13
8. Compiling and loading models	14
9. Statistics, yield and optimization	14
10. Post-layout and reliability	15
11. Interactive debugging and control	15
12. Control-flow scripting (.control)	16
13. Help and utility commands	16

ngspice commands — a worked reference

This document walks through the commands available in this ngspice build — the stock nutmeg/spice command set plus every command added by the [enhancement project](#) (the RF steady-state suite, the optimizer, Monte-Carlo/yield, post-layout reliability, pyplot, pre_osdi/pre_snp, eye, stb, ...). The core analyses are shown with complete, copy-pasteable netlists and plots produced by actually running them; the many smaller utility commands are collected into reference tables at the end.

Every plot in this document was generated by [make_commands_figs.py](#) from the exact netlists shown.

1. How commands are run

An ngspice input file (a *deck* or *netlist*) has this shape — the first line is always the title and is ignored as a component:

```
* my circuit title          <- title line (ignored)
V1 in 0 dc 1                <- components
R1 in out 1k
C1 out 0 1u
.tran 10u 5m                <- a "dot-card" analysis
.control                    <- an interactive command block
run
```

```
plot v(out)
.endc
.end
```

There are four ways a command runs:

Context	How	Example
Batch	ngspice -b deck.cir runs every dot-card analysis and any .control block, then exits	ngspice -b amp.cir
Interactive	at the ngspice N -> prompt	ngspice -> tran 1u 1m run then print v(out)
.control block	commands between .control and .endc in the deck	.ac dec 10 1 1meg
Dot-card	.tran/.ac/... in the deck body, executed by run	

The interactive analysis commands (tran, ac, dc, op, ...) take the same arguments as their dot-card spelling and run immediately. run executes the analysis named by the deck's dot-cards. Results are stored as named vectors inside plots (print, plot, and expressions read them back).

2. Analyses

op — DC operating point

Solves the quiescent (DC) bias point: every node voltage and source current with all capacitors open and inductors shorted.

```
* operating point of a resistive divider
V1 in 0 dc 5
R1 in out 2k
R2 out 0 3k
.control
op
print v(out) i(v1)
.endc
.end
```

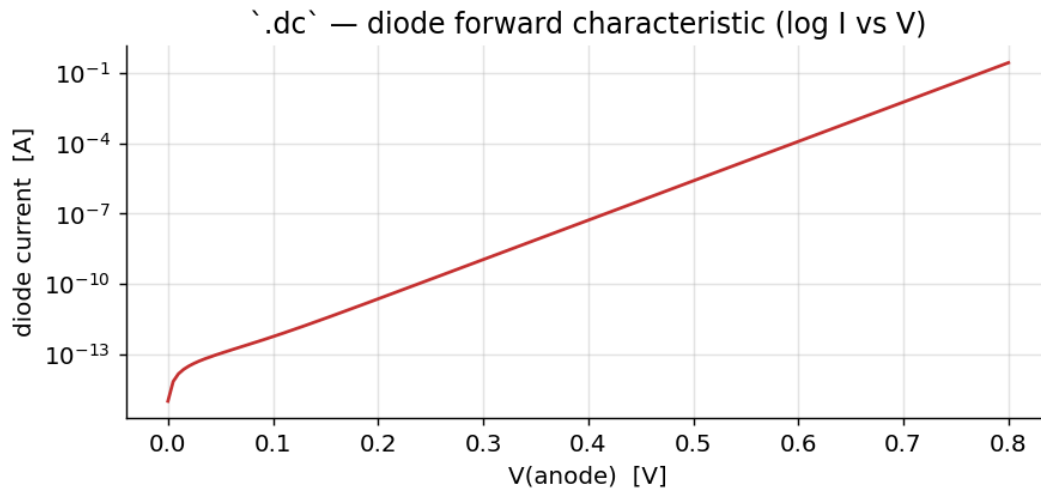
v(out) = 3.0 V, i(v1) = -1 mA. op is also the bias point every AC, noise, tf, pz, and small-signal analysis linearizes around.

dc — DC sweep (and nested sweep)

dc <src> <start> <stop> <step> [<src2> <start2> <stop2> <step2>] sweeps a source (and optionally a second, giving a family of curves). The classic uses are a diode I-V and a transistor's output characteristics.

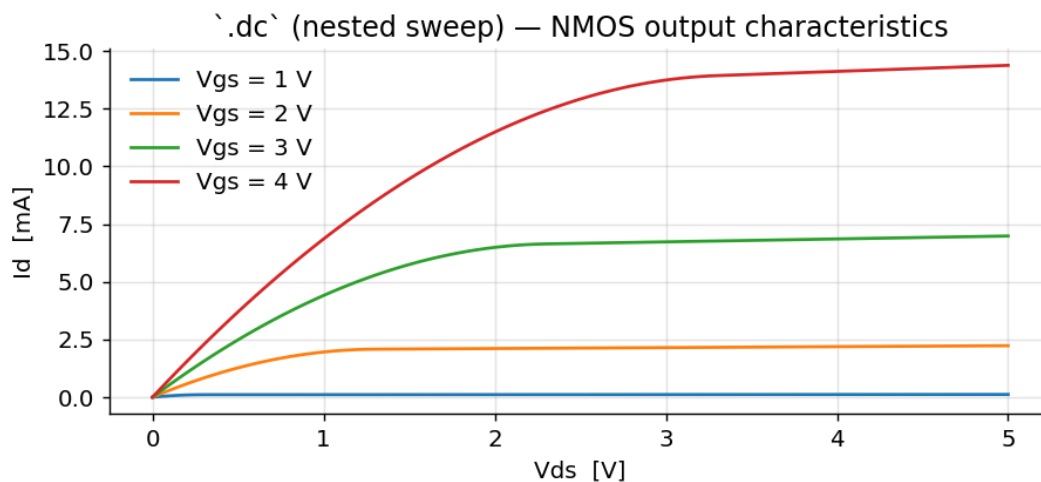
```
* diode forward I-V
Vd a 0 dc 0
D1 a 0 DMOD
.model DMOD D(is=1e-14 n=1)
.dc Vd 0 0.8 0.005
.control
run
```

```
plot abs(i(vd)) ylog
.endc
.end
```



A nested sweep produces the textbook MOSFET output family — the inner source V_{ds} sweeps for each value of the outer source V_{gs} :

```
* NMOS output characteristics (nested .dc)
Vds d 0 dc 0
Vgs g 0 dc 0
M1 d g 0 0 NM W=20u L=1u
.model NM NMOS(level=1 vto=0.7 kp=120u lambda=0.02)
.dc Vds 0 5 0.05 Vgs 1 4 1
.control
run
plot -i(vds)
.endc
.end
```



ac — small-signal frequency response

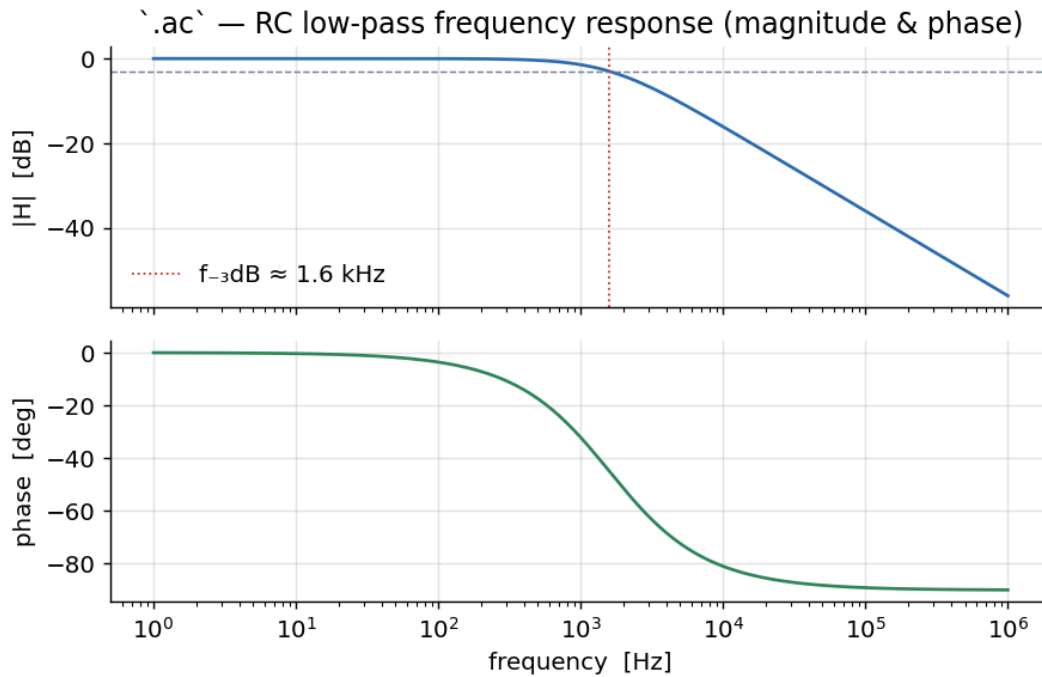
`ac <lin|dec|oct> <N> <fstart> <fstop>` linearizes at the operating point and sweeps frequency. Drive the input with an ac magnitude and read `vdb()` (magnitude in dB) and `vp()` (phase).

* RC low-pass Bode

```

Vin in 0 dc 0 ac 1
R1 in out 1k
C1 out 0 100n
.ac dec 40 1 1meg
.control
run
plot vdb(out)
plot vp(out)
.endc
.end

```



The -3 dB corner sits at $1/(2\pi RC) \approx 1.6$ kHz; the phase rolls from 0° to -90° .

tran — transient (time-domain)

`tran <tstep> <tstop> [tstart [tmax]] [uic]` integrates the circuit in time.

* 1 V step into an RC (over-damped) and a series RLC (ringing)

```
Vin in 0 pulse(0 1 1u 1n 1n 1 2)
```

```
R1 in a 1k
```

```
C1 a 0 100n
```

```
L1 in b 1m
```

```
Rs b c 20
```

```
C2 c 0 100n
```

```
.tran 1u 400u
```

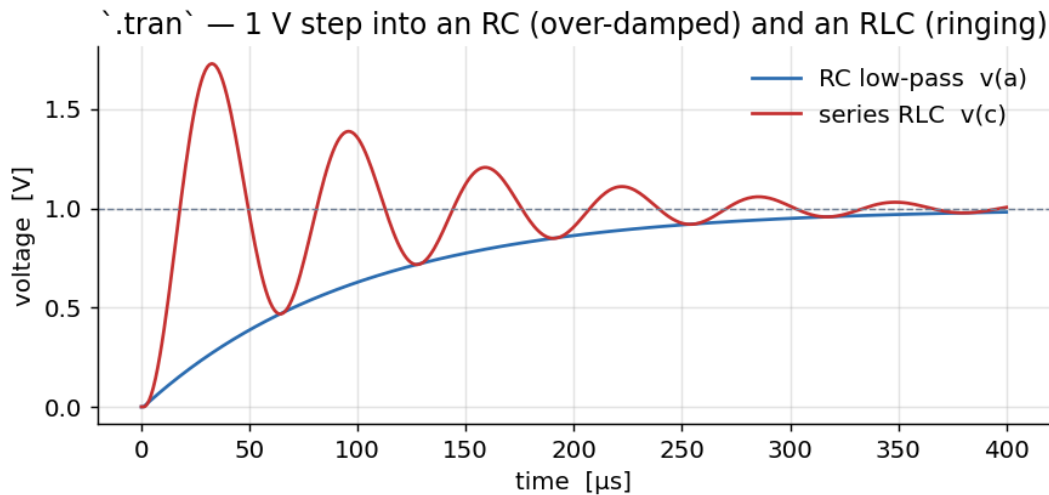
```
.control
```

```
run
```

```
plot v(a) v(c)
```

```
.endc
```

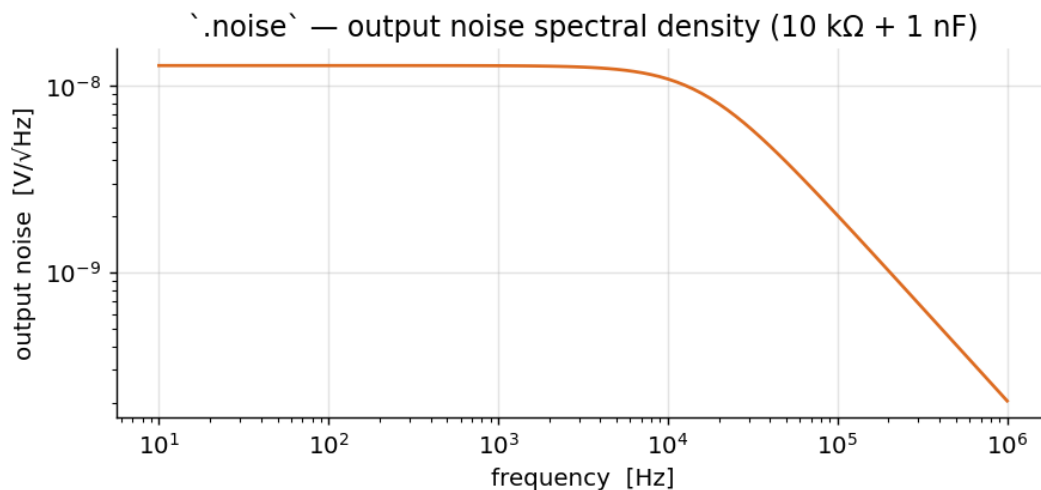
```
.end
```



noise — noise analysis

`noise <output> <input-source> <lin|dec|oct> <N> <fstart> <fstop>` computes the output- and input-referred noise spectral density (and integrated noise). The spectrum vectors live in the `noise1` plot; the integrated totals in `noise2`.

```
* output noise of a 10k + 1n RC
Vin in 0 dc 0 ac 1
R1 in out 10k
C1 out 0 1n
.noise v(out) Vin dec 20 10 1meg
.control
run
setplot noise1
plot onoise_spectrum
.endc
.end
```



tf — DC transfer function

`tf <output> <input-source>` reports the small-signal DC gain plus the input and output impedance. `<output>` is `v(node)`, `v(node1,node2)`, or `i(vsource)`.

```
* transfer function of a voltage divider
```

```

Vin in 0 dc 1
R1 in out 1k
R2 out 0 1k
.control
tf v(out) Vin
.endc
.end

```

Reports transfer_function = 0.5, output_impedance_at_v(out) = 500 Ω , vin#input_impedance = 2 k Ω .

pz — pole/zero

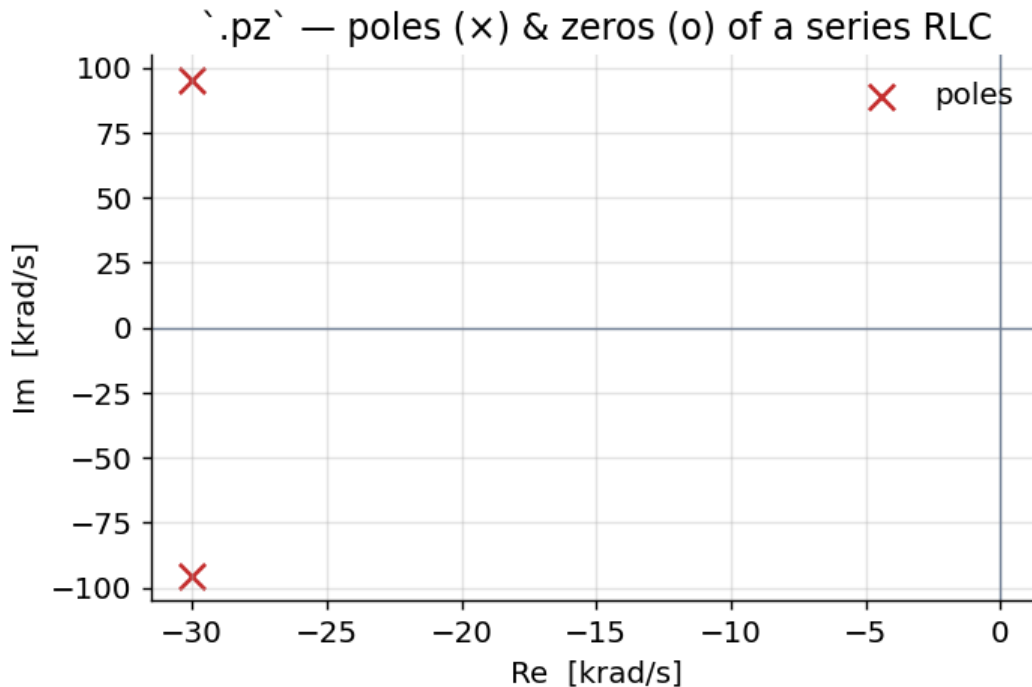
pz <n1> <n2> <n3> <n4> <cur|vol> <pz|pol|zer> finds the poles and zeros of the transfer function between the input node pair (n1,n2) and output pair (n3,n4).

* poles/zeros of a series RLC (v(out)/vin)

```

Vin in 0 dc 0 ac 1
L1 in m 1m
R1 m out 60
C1 out 0 100n
.pz in 0 out 0 vol pz
.control
run
print all
.endc
.end

```



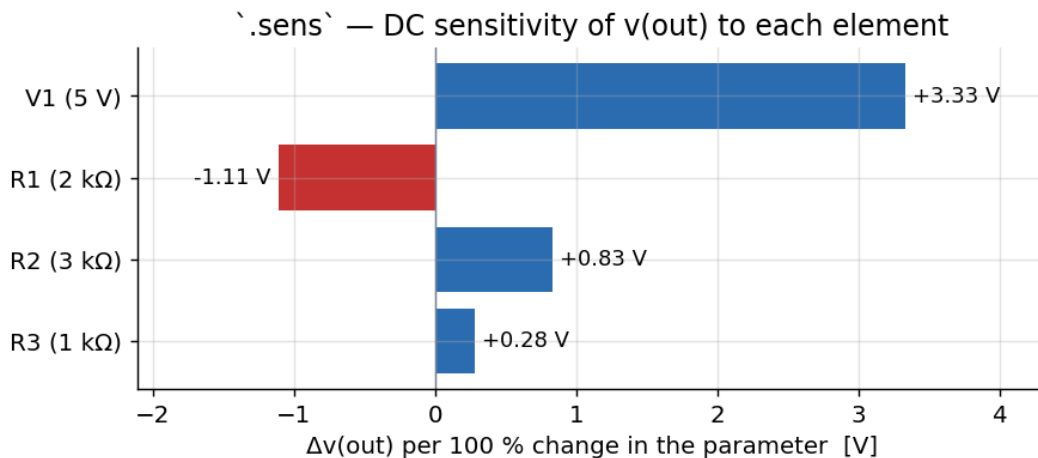
The complex-conjugate pole pair sits at -30 ± 95 krad/s — a resonance near $\omega_0 = 1/\sqrt{LC} = 100$ krad/s with the loss set by R.

sens — sensitivity

sens <output> computes the DC sensitivity of an output to every circuit parameter; append **ac** <lin|dec|oct> <N> <fstart> <fstop> for the small-signal AC sensitivity vs frequency. The result lands in a **sens1** plot, one vector per device (its value sensitivity) and per model parameter (<dev>: <param>).

```
* DC sensitivity of v(out) to every element
V1 in 0 dc 5
R1 in out 2k
R2 out mid 3k
R3 mid 0 1k
.sens v(out)
.control
run
print v1 r1 r2 r3      ; each is  $\partial v(\text{out})/\partial \text{param}$ 
.endc
.end
```

Scaling each raw sensitivity by the parameter's nominal value gives a comparable “ $\Delta v(\text{out})$ per 100 % change” — the clearest way to rank which element matters most:



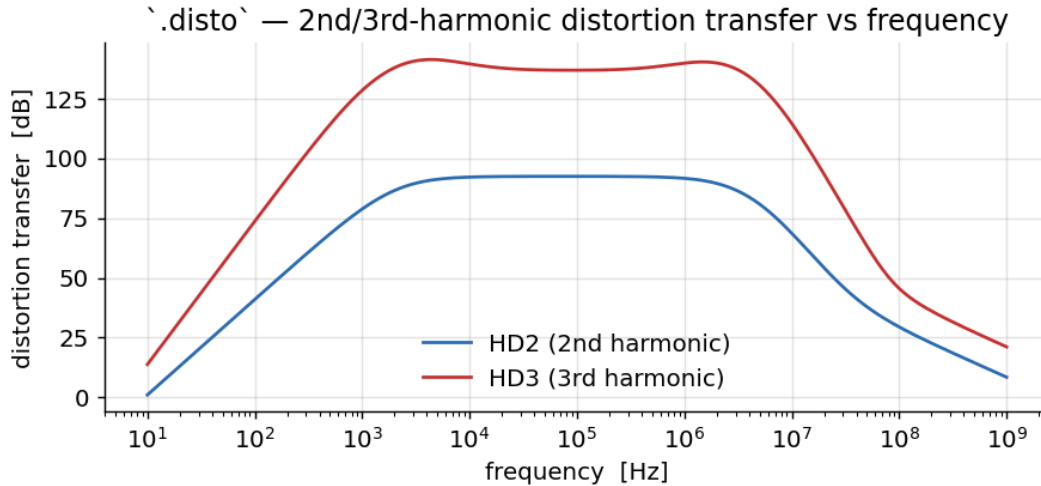
The output tracks the 5 V supply most strongly, and increasing the series R_1 *lowers* it (negative sensitivity), as expected for a divider.

disto — small-signal distortion

disto <lin|dec|oct> <N> <fstart> <fstop> [f2overf1] sweeps a small-signal harmonic (and, with a second tone, intermodulation) distortion analysis. Mark the driving source with **distof1** (and **distof2** for two-tone). The 2nd- and 3rd-harmonic products land in the **disto1** / **disto2** plots.

```
* small-signal distortion of a common-emitter BJT amp
Vcc cc 0 dc 5
Vin in 0 dc 0.75 ac 1 distof1 1
Rb cc b 220k
Rc cc c 2.2k
Cin in b 1u
Cout c 0 20p
Q1 c b 0 QMOD
.model QMOD NPN(is=1e-15 bf=150 vaf=80 cjc=3p tf=0.3n)
.disto dec 20 10 1g
```

```
.control
run
setplot disto1
plot vdb(c)          ; HD2 vs frequency (disto2 holds HD3)
.endc
.end
```



The distortion (here in dB relative to the unit `distof1` drive) is flat across the amplifier's passband and rolls off with the input coupling capacitor at low frequency and the transistor's bandwidth at high frequency.

fourier

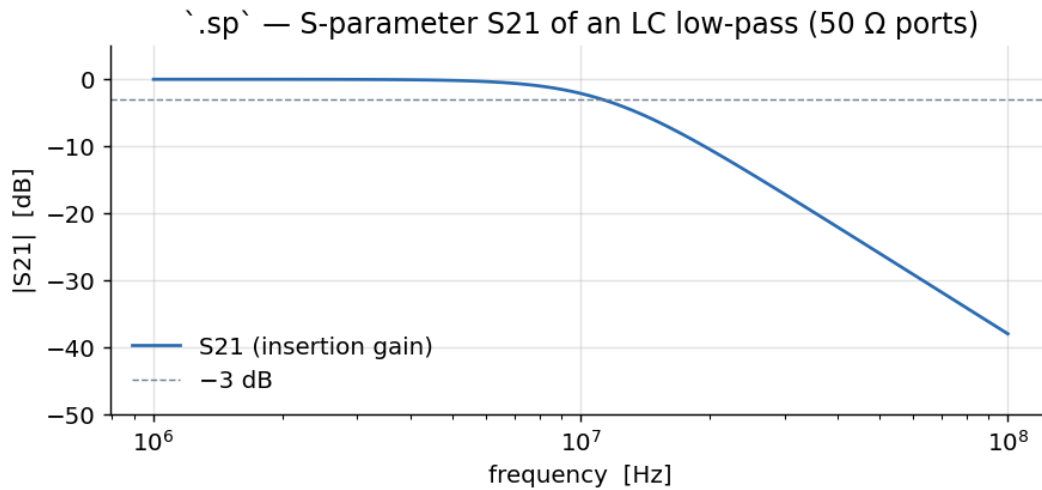
`fourier <fund_freq> <vec>` prints the DC term, the first harmonics, and the THD of a time-domain waveform — run a `.tran` first, then `fourier 10k v(out)`. (The `spec/fft` commands in §4 give the full spectrum as a plot.)

3. RF and periodic steady-state

The project adds a full RF suite on top of the analyses above. Most run as their own command (or dot-card); the periodic small-signal analyses (`.pac`, `.pnoise`, `.pxf`, `.psp`) run as dot-cards around a `pss/.pss` operating point.

`sp` is the small-signal S-parameter analysis — tag the ports with `portnum/z0` sources:

```
* LC low-pass S-parameters (50 ohm ports)
L1 in out 1u
C1 out 0 400p
V1 in 0 dc 0 ac 1 portnum 1 z0 50
V2 out 0 dc 0 ac 1 portnum 2 z0 50
.sp dec 50 1meg 100meg
.control
run
plot vdb(S_2_1)
.endc
.end
```

Command	What it does
<code>sp</code>	S-parameters + noise figure (<code>.sp</code>), Touchstone-exportable
<code>pss</code>	periodic steady state (shooting); the base for the periodic small-signal set
<code>hb</code>	harmonic-balance steady-state spectrum: <code>hb <f0> <K></code>
<code>hbosc</code>	autonomous HB for an oscillator: <code>hbosc <oscnode> <K> [fguess] [tstab]</code>
<code>phasenoise</code>	oscillator phase-noise $L(\Delta f)$ around an <code>hbosc</code> point
<code>qpss</code>	two-tone quasi-periodic steady state: <code>qpss <expr> <f1> <f2> ...</code>
<code>qpac / qpxf / qpnoise</code>	two-tone periodic AC / transfer / noise around a <code>qpss...hb</code> point
<code>envelope</code>	envelope following: <code>envelope <node> <fc> <tstop></code>
<code>stb</code>	loop-gain stability (Middlebrook/Tian double injection); reports phase/gain margin

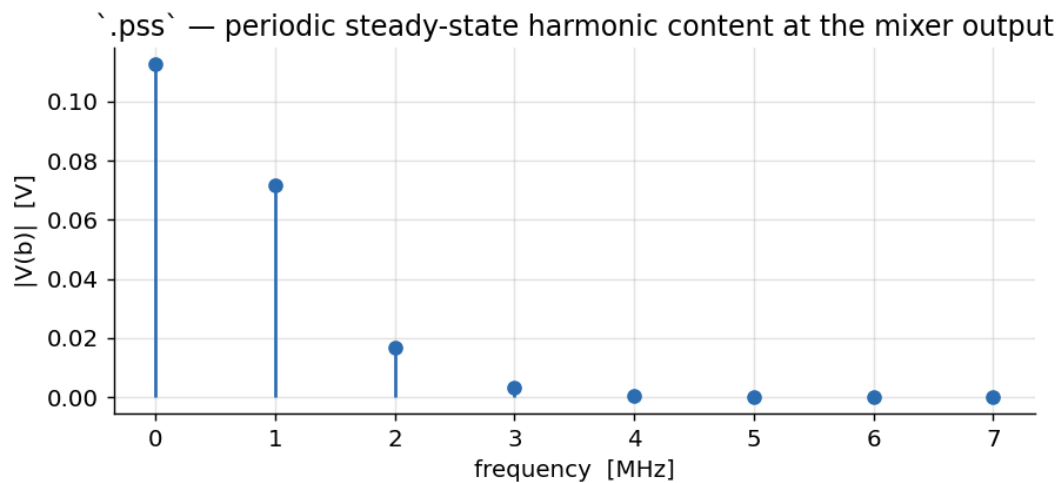
Periodic small-signal — `.pss`, `.pac`, `.pnoise`, `.pxf`, `.psp`

These analyses characterise a circuit around a large-signal periodic operating point (a mixer under its LO, a switched-capacitor under its clock, an oscillator). `.pss` finds that periodic steady state; the rest are small-signal analyses *on top of it*. All share the same leading arguments `<fund>` `<tstab>` `<outnode>` `<points>` `<harmonics>`

A self-contained example — a behavioural (B-source) conductance pumped by a 1 MHz "LO", which acts as a mixer:

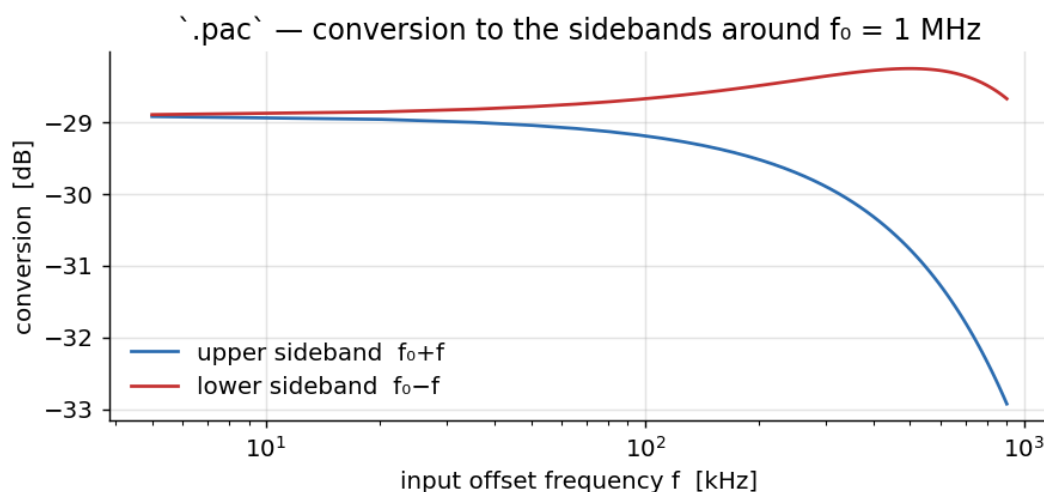
```
* LO-pumped behavioural mixer
V1 lo 0 SIN(0 1 1meg)
VDC a 0 DC 1 AC 1
R1 a b 10k
B1 b 0 I=(1m + 0.8m*v(lo))*v(b)      ; conductance modulated by the LO
C1 b 0 100p
.pss 1meg 1u b 1024 8 50 5u          ; find the periodic steady state
.control
run
setplot pss2                          ; frequency-domain result
plot mag(b)
.endc
.end
```

`.pss` returns the periodic operating point — its harmonic content shows the LO and the harmonics the nonlinearity generates:



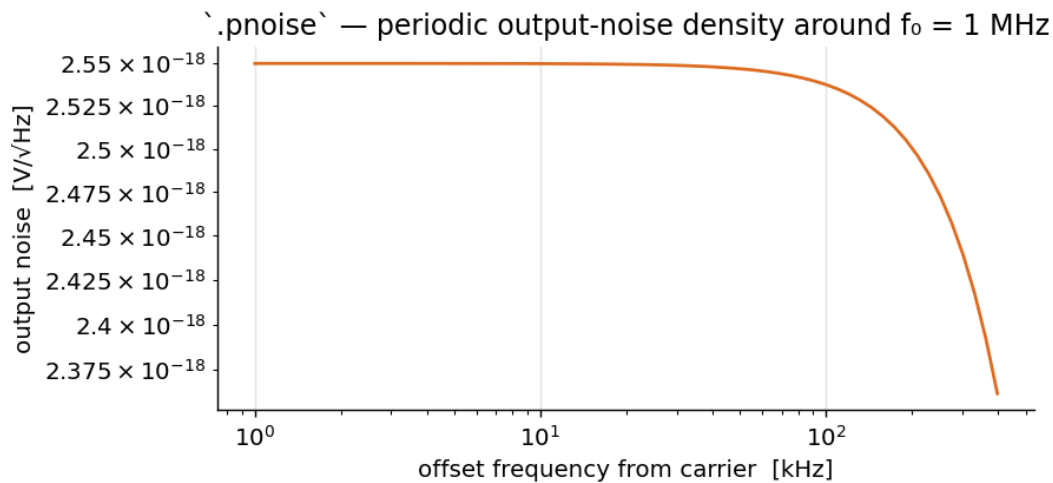
With that operating point, `.pac` computes the small-signal conversion to each mixing sideband ($f_0 \pm f$) versus the input frequency — swap the `.pss` line for:

```
.pac 1meg 1u b 1024 6 50 5u lin 60 5k 900k 1
plot db(b_usb1) db(b_lsb1)      ; upper / lower sideband conversion
```



`.pnoise` gives the periodic / phase-noise density folded from every sideband, versus offset from the carrier:

```
.pnoise 1meg 1u b 1024 6 50 5u b vdc dec 25 1k 400k
set sqnoise
plot onoise_spectrum          ; V/√Hz vs offset frequency
```



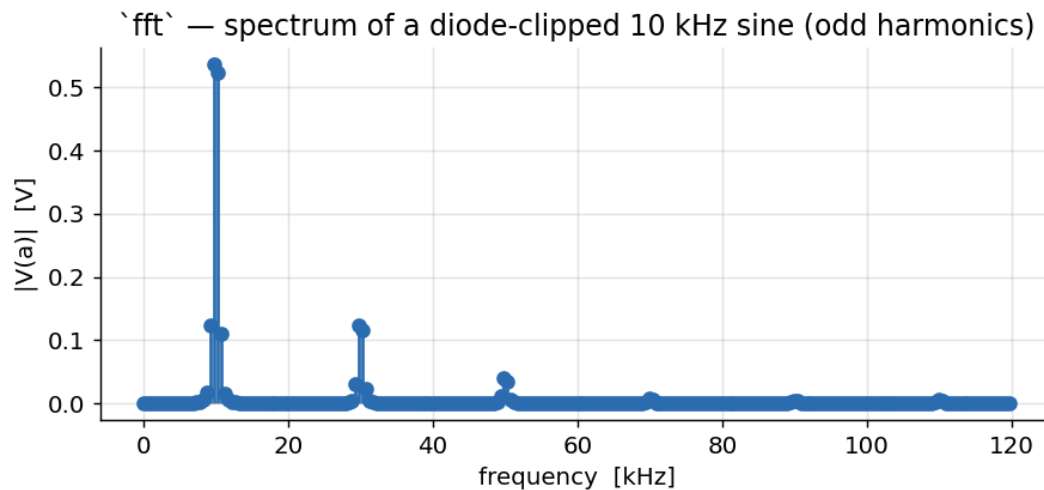
`.pxf` (periodic transfer function — sensitivity of an output to a small perturbation at each sideband) and `.psp` (periodic S-parameters) take the same `.pss` preamble; see the [RF suite reference](#) for those plus the two-tone `qpss/qpac/qnoise/qpxf` family and the oscillator `hbosc/phasenoise` flow.

4. Signal processing and measurement

These operate on vectors already in memory (typically after a `.tran`).

`fft <vec>` computes an FFT (run `linearize` first so the samples are uniformly spaced); `spec` and `psd` are related spectral views.

```
* harmonics of a diode-clipped 10 kHz sine
Vin in 0 sin(0 1.2 10k)
R1 in a 1k
D1 a 0 DL
D2 0 a DL
.model DL D(is=1e-12)
.tran 1u 2m
.control
run
linearize
fft v(a)
plot mag(v(a))
.endc
.end
```



The symmetric clipper produces only odd harmonics (30, 50, 70 kHz ...).

Command	Purpose
meas	measure a quantity from a result (meas tran tr TRIG ... TARG ..., meas ac gain MAX vdb(out), gain/phase margin, delays, RMS, ...)
fft	FFT spectrum of a vector
spec	spectrum over a frequency window: spec <fstart> <fstop> <fstep> <vec>
psd	power spectral density via FFT
fourier	Fourier coefficients + THD of a transient
eye	eye diagram + jitter: eye <expr> -ui <T> [-tstart t0] [-threshold v]
linearize	resample a transient onto a uniform time grid (needed before fft)
cutout	keep only a slice of a vector
transpose	transpose a multi-dimensional vector
cross	build a vector by picking elements from others
compose	build a vector from values/ranges: compose v start=0 stop=1 step=0.1
reshape	change a vector's dimensions
settype / deftype	set/redefine a vector's type (voltage, time, ...) for axis labelling

5. Vectors, variables and expressions

Results are vectors; `let` creates or modifies them and `print/plot` read them. Expressions support the usual math plus `v()`, `i()`, `vdb()`, `vp()`, `mag()`, `ph()`, `real()`, `imag()`, `db()`, `abs()`, `sqrt()`, `vector()`, `length()`, `mean()`, `vecmax()`, `vecmin()`, and the ternary `?:`.

```
.control
run
let pwr = v(out)*i(vout)           ; a new vector
let gain_db = vdb(out)
print vecmax(abs(v(out)))           ; scalars are 1-length vectors
print mean(pwr)
.endc
```

Command	Purpose
let	assign a vector: let x = expr

Command	Purpose
unlet	delete vectors
print	print vector values: print [col] expr ...
display	list the vectors in the current plot
define / undefine	user function: define sq(x) x*x
set / unset	set/clear a control variable: set numdgt=8
setcs	like set, but keeps the value's case
option / options	set a simulator option: option reltol=1e-4 klu
setscale	change the plot's default scale vector
setseed	reseed the RNG for reproducible Monte Carlo
remzerovec	drop zero-length vectors

6. Plotting and data export

```
.control
run
plot v(out) vs time
plot vdb(out) xlog          ; log-x Bode
pyplot acmag vdb(out)      ; matplotlib PNG/-window
wrdata out.dat v(out) v(in) ; columnar data file
write out.raw v(out)       ; binary rawfile (reload with `load`)
.endc
```

Command	Purpose
plot	the built-in plotter: plot expr ... [vs expr] [xl lo hi] [yl lo hi] [xlog] [ylog]
pyplot	render/export via matplotlib (-eye, -hist, -contour variants); PNG when a file is named
gnuplot	render/export via gnuplot
hardcopy	write a PostScript/SVG hardcopy of a plot
asciiplot	plot as ASCII art in the terminal
wrdata	write columnar (whitespace) data — each vector prefixed by its scale
write	write a binary rawfile (load reads it back)
load	load a rawfile into a new plot
wrsnp / rdsnp /	Touchstone .sNp write / read / 2-port write
wrs2p	
diff	compare two plots vector-by-vector
setplot	switch the current working plot
destroy	free a plot's data

7. Circuit and device management

```
.control
source amp.cir          ; read a deck
listing                 ; show the parsed circuit
alter r1 = 2k           ; change a device value, then re-run
altermod nmos vth0=0.4  ; change a .model parameter
alterparam vsup=3.3     ; change a .param, then `reset` to apply
```

```
show m1          ; print device operating-point parameters
run
.endc
```

Command	Purpose
source	read and load a netlist file
listing	print the current circuit (logical/physical/deck)
edit	edit the deck in \$EDITOR, then reload
cirbyline	feed the circuit one line at a time (scripting)
alter / altermod / alterparam	change device / .model / .param values
show / showmod	print device / model parameters (operating point)
devhelp	list device types and their parameters
inventory	count the devices in the circuit
save	select which node/branch outputs to keep
setcirc	switch between loaded circuits
remcirc / removecirc	drop the current circuit
reset	rebuild the circuit (apply alterparam, clear state)
dump	dump the circuit matrix structure

8. Compiling and loading models

This build compiles Verilog-A to OSDI with openvaf-r and loads it directly.

```
.control
pre_osdi bsim4.osdi          ; load a compiled Verilog-A model (the `osdi` command)
pre_osdi -f bsim4.osdi      ; force-reload after recompiling (no restart)
pre_snp filter.s2p          ; compile a Touchstone file to an OSDI n-port, then load it
codemodel extra.cm          ; load an XSPICE code-model library
.endc
N1 d g s b bsim4mod          ; an OSDI device instance (N-prefix)
.model bsim4mod bsim4va w=10u l=1u
```

Command	Purpose
osdi (pre_ osdi)	load one/more compiled .osdi models; -f force-reloads a recompiled file
snp (pre_snp)	compile a Touchstone .sNp to a Verilog-A n-port OSDI model, then load it
codemodel	load an XSPICE code-model (.cm) library
use	load a device library

9. Statistics, yield and optimization

```
* Monte-Carlo yield of an RC corner
.param rr=agauss(1k, 5p, 3)      ; 5% (3-sigma) Gaussian on R
R1 in out {rr}
C1 out 0 1n
Vin in 0 ac 1
.ac dec 20 1 1meg
.control
```

```
montecarlo 500 -analysis ac -spec 'vdb(out)' -min -3.5 -max -2.5
.endc
.end
```

Command	Purpose
montecarlo	packaged yield: N samples, per-spec pass/fail, yield + 95% CI
mcsample	Latin-Hypercube sampling of the .param random draws (lower variance)
highsigma	rare-event / high-sigma failure probability by scaled-sigma importance sampling
mccorr	register a correlation matrix; draw with mvnorm() in .param
setseed	reproducible random draws
sweep	sweep any knob (device/model/.param) and record outputs into a plot
optimize	Nelder-Mead / Levenberg-Marquardt parameter optimizer against an objective

See the [statistics](#) and [optimizer](#) references for full treatments with plots.

10. Post-layout and reliability

Command	Purpose
reduce	reduce an R/C parasitic network (TICER) to an equivalent .subckt valid to fmax
aging	age aging-capable devices to a target lifetime (HCI/NBTI/TDDDB) for fresh-vs-aged runs
emir	power-grid IR-drop + electromigration current density + Black's-equation MTTF

11. Interactive debugging and control

```
.control
stop when v(out) > 0.9      ; breakpoint on a condition
trace v(out)                ; watch a node
run                          ; runs until the breakpoint
where                        ; last non-converging node/device
resume                      ; continue
savestate ckpt.st           ; checkpoint the transient state to disk
* ... later, even in a fresh process ...
loadstate ckpt.st           ; restore and continue the .tran
.endc
```

Command	Purpose
run	run the analysis from the deck's dot-cards (optionally to a rawfile)
resume	continue after a breakpoint/stop
stop	set a breakpoint (stop after N, stop when expr)
step	advance N timepoints
trace / iplot	watch / incrementally plot nodes during a run
status / delete	list / remove breakpoints and traces
savestate /	checkpoint / restore the full transient state (cross-process)
loadstate	
snsave / snload	save / load a solution snapshot
where	print the last non-converging node or device

12. Control-flow scripting (**.control**)

.control blocks are a small shell-like language. In addition to the commands above:

```
.control
run
* structured control
if v(out) > 0.5
  echo "high"
else
  echo "low"
end
* loops
foreach f 1k 10k 100k
  ac dec 10 $f $f
  print vdb(out)
end
let k = 0
dowhile k < 3
  echo iteration $k
  let k = k + 1
end
.endc
```

Construct	Purpose
if / else / end	conditional
while / dowhile / end	pre-/post-test loops
foreach var v1 v2 ... / end	iterate over a value list
repeat [N] / end	repeat N times (or forever)
break / continue	loop control
label / goto	jump within a block
shift	shift a list/argv variable left
echo	print text/\$variables
shell	run a host shell command
cd / getcwd	change / print working directory
strcmp / strstr / strslice	string compare / search / slice into \$var
fopen / fread / fclose	open / read / close a file into shell variables

13. Help and utility commands

Command	Purpose
help / tutorial / newhelp / oldhelp	documentation browser (help <command>)
history	recall previous commands (history [-r] [N])
alias / unalias	define / remove a command alias
synhl	preview interactive syntax highlighting of a line
version	print the version
sysinfo	print host CPU/memory summary
rusage	print resource usage (time, memory, all)
cdump	dump the parsed control structures
mdump / mrdump	dump the matrix / RHS (debugging)

Command	Purpose
bug	open a bug report
rehash	rebuild the host-command hash
quit	exit ngspice

Developer/edge commands: aspice/rspice/jobs (async/remote runs), mc_source, optran, wrnodev, check_ifparm, bltplot (a plot alias), state (unimplemented placeholder).

Generated for the ngspice-46 + OpenVAF-r enhancement project. Plots produced by make_commands_figs.py from real ngspice runs; rebuild the PDF with build_commands_pdf.py.